

Overview

Part 02 —  Deep Learning · 02-deep-learning/README.md

Module 02 — Deep Learning

Where you are: Stop 2 of 16. **Time:** ~25–30 hours · **Prereq:** Module 01 [01-ml-foundations].

One neuron is logistic regression. Stack them, learn the intermediate representations instead of hand-crafting them, and you get deep learning. This module is where you stop *choosing* features and start *learning* them — the shift that makes everything from Module 03 onward possible.

Learning objectives

1. Implement a multi-layer perceptron and **backpropagation** from scratch, and verify it with a gradient check.
2. Explain vanishing/exploding gradients and the three inventions that fixed them (ReLU, normalization, residuals).
3. Build and train networks in PyTorch: `Dataset` → `DataLoader` → `Module` → training loop → checkpoint.
4. Choose between MLP / CNN / RNN by the *structure of the data*, not by fashion.
5. Diagnose a failing training run in under 15 minutes using a fixed checklist.
6. Apply transfer learning and fine-tuning — the workflow you'll reuse for LLMs in Module 04.

Suggested path

#	Do this	File	Time
1	Core concepts	notes/01-neural-networks.md	4h
2	Training craft	notes/02-training-deep-networks.md	4h
3	Backprop, by hand	code/backprop_from_scratch.py	3h
4	PyTorch in one file	code/pytorch_training_loop.py	3h
5	Slides	slides/	1h
6	Lab	lab/EXERCISES.md	8h
7	Papers	papers/PAPERS.md	5h

The 12 terms you must own

neuron · activation · backpropagation · computational graph · vanishing gradient · batch normalization · residual connection · dropout · embedding · convolution · transfer learning · mixed precision

Exit check

You can train a network on a new dataset, hit a target metric, explain every line of your training loop, and debug it when the loss goes to NaN — without copying a notebook.